



**Institute of Information Technology
Jahangirnagar University**

AI-Based Paddy Disease Detection and Forecasting System
Prototype Project Proposal

Submitted By

Md. Shaon Khan
Tahshin Islam Tuhin
Md. Ammar Hamim

Supervisor

Dr. Risala Tasin Khan
Professor
Institute of Information Technology

Institute of Information Technology
18 May 2026

AI-Based Paddy Disease Detection and Forecasting System

Prototype Project Proposal

Introduction

Agriculture plays a major role in the economy and food production of many countries. Among different crops, paddy is one of the most important agricultural products. However, paddy production is heavily affected by various diseases caused by bacteria, viruses, fungi, environmental stress, and poor soil conditions. In many cases, farmers are unable to identify diseases at an early stage, which leads to reduced crop quality and significant production loss.

Traditional disease identification methods mainly depend on manual observation and agricultural experts. These methods are often slow, expensive, and not always available in rural areas. With the advancement of Artificial Intelligence, Machine Learning, and Deep Learning technologies, it is now possible to develop intelligent systems that can assist farmers by automatically detecting diseases and predicting future risks.

This project proposes an AI-based smart agriculture system capable of analyzing paddy leaf images, weather conditions, soil information, and crop health conditions to detect existing diseases and forecast possible future diseases. The system will also provide preventive suggestions and recommendations to support better farming decisions.

Project Goal

The main goal of this project is to develop an intelligent agricultural support system that can detect paddy diseases and predict future disease risks using image analysis and environmental data.

The system aims to assist farmers by providing early warnings, disease analysis, and preventive recommendations to reduce crop damage and improve agricultural productivity.

Objectives

Primary Objectives

- Detect paddy diseases from crop images
- Predict possible future diseases using environmental conditions
- Analyze soil and weather conditions
- Provide preventive suggestions and treatment recommendations
- Assist farmers in early decision making

Secondary Objectives

- Build a smart agriculture monitoring system
- Reduce crop loss caused by delayed disease detection
- Improve disease awareness among farmers
- Support sustainable and efficient farming methods
- Create a scalable AI-based agricultural platform

Problem Statement

Paddy diseases spread rapidly under favorable environmental conditions such as high humidity, excessive rainfall, poor soil condition, and temperature changes. Farmers often fail to recognize symptoms at the early stage, which causes disease outbreaks and severe crop damage.

Existing agricultural support systems are limited because many of them only focus on image classification and ignore environmental and soil-related factors. However, disease development depends not only on visible symptoms but also on future weather conditions, soil quality, and crop health status.

Therefore, there is a need for an intelligent system that can combine image analysis, weather forecasting, soil monitoring, and predictive analytics to provide a complete agricultural disease management solution.

Proposed System Overview

The proposed system will work as a complete AI-powered agricultural assistant. The system will collect and analyze multiple types of information including crop images, soil data, weather conditions, and crop health information.

Initially, the system will determine whether the uploaded sample belongs to a normal or diseased category. If the sample is diseased, the system will further classify the disease into broader categories such as bacterial, viral, fungal, or other unknown conditions. After identifying the disease category, the system will perform another level of classification to identify the specific disease name.

In addition to disease detection, the system will analyze historical weather data, current environmental conditions, upcoming 10 to 15 day weather forecasts, soil conditions, and crop health information to predict possible future disease risks.

The system will finally generate recommendations, warning alerts, and preventive suggestions for farmers.

Disease Classification Structure

Level 1 Classification

The system will first classify the crop condition into:

- Normal
- Diseased
- Unknown/Unrecognized Condition

Level 2 Classification

If the sample is diseased, the system will classify it into:

- Bacterial Disease
- Viral Disease
- Fungal Disease
- Other Disease Types

Level 3 Classification

After identifying the disease category, the system will identify the specific disease name.

Examples include:

Bacterial Diseases

- Bacterial Leaf Blight
- Bacterial Leaf Streak

Viral Diseases

- Rice Tungro Virus
- Rice Ragged Stunt Virus

Fungal Diseases

- Rice Blast
- Brown Spot

- Sheath Blight

System Inputs

Image Inputs

- Paddy leaf image
- Stem image
- Root image
- Grain image (optional)

Weather Inputs

- Temperature
- Humidity
- Rainfall
- Wind speed
- Seasonal condition

Soil Inputs

- Soil moisture
- Soil pH
- Nutrient condition
- Soil temperature

Crop Health Inputs

- Current crop condition
- Infection severity
- Growth stage

System Outputs

The system will generate the following outputs:

- Disease category
- Specific disease name
- Confidence score
- Disease severity level
- Future disease risk
- Warning alerts
- Preventive recommendations
- Natural remedy suggestions

Future Disease Prediction Concept

Disease development depends strongly on environmental conditions. Certain diseases spread rapidly under specific weather and soil situations.

For example:

- High humidity may increase fungal and bacterial infections
- Heavy rainfall may increase leaf infection risk
- Hot temperatures may support viral spread
- Poor soil conditions may weaken crop resistance

The system will analyze historical environmental patterns, current crop condition, and upcoming weather forecasts to estimate possible disease risks within the next 10 to 15 days.

Recommendation System

After disease analysis and forecasting, the system will generate recommendations for farmers.

Possible recommendations include:

- Proper irrigation management
- Isolation of infected plants
- Biological pest control methods
- Organic treatment suggestions
- Soil management guidance
- Preventive monitoring alerts

The recommendation system may also include natural and eco-friendly farming suggestions.

Proposed Workflow

Step 1

Farmer uploads crop image.

Step 2

System collects weather and soil information.

Step 3

Image preprocessing and environmental data preprocessing are performed.

Step 4

The system detects whether the crop is healthy or diseased.

Step 5

Disease category classification is performed.

Step 6

Specific disease identification is performed.

Step 7

Weather and soil analysis are combined with crop health status.

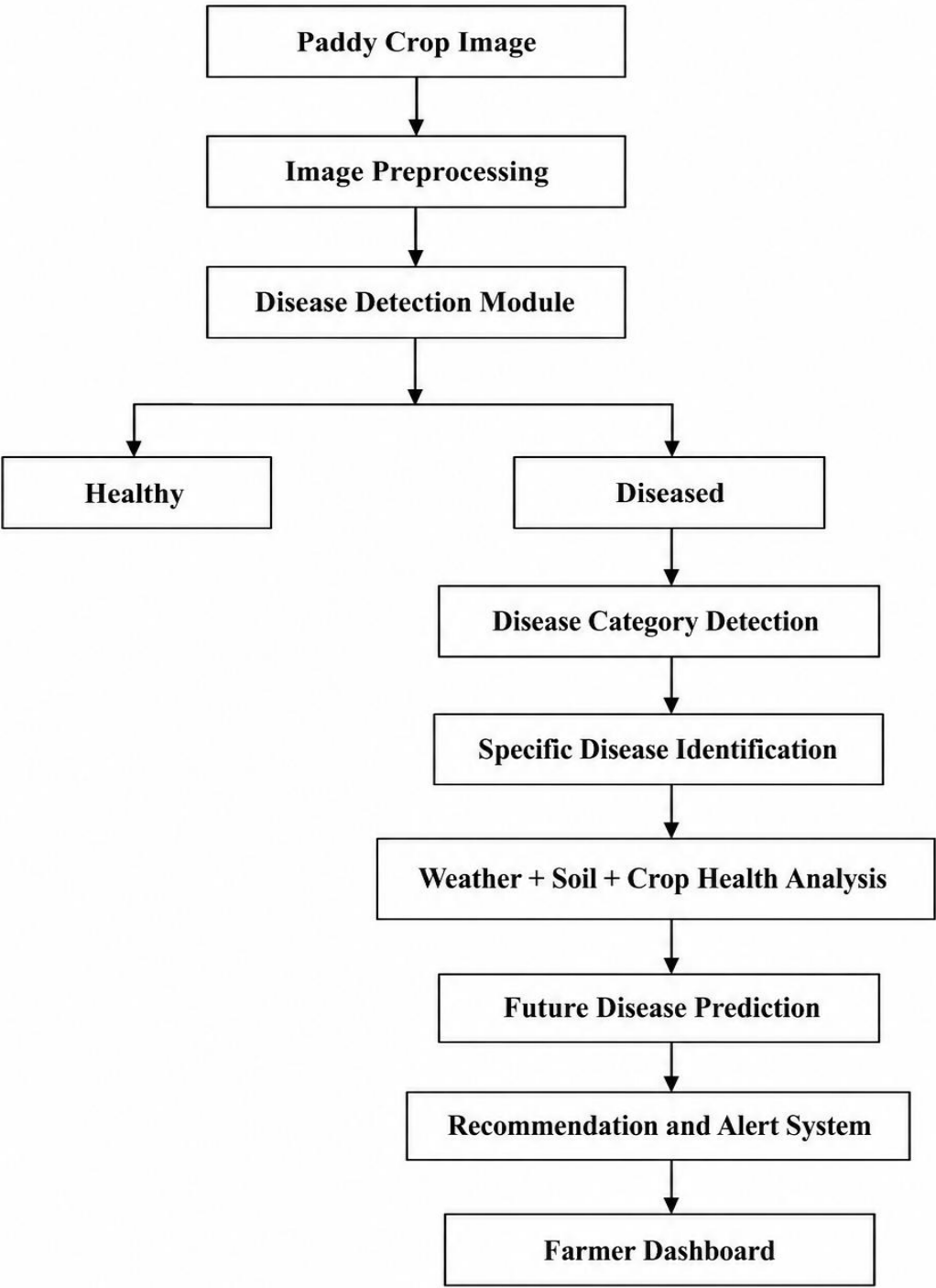
Step 8

Future disease risk prediction is generated.

Step 9

Recommendations and warning alerts are provided to the farmer.

Workflow Diagram



Possible Technologies

Purpose	Possible Technology
Deep Learning	TensorFlow / PyTorch
Backend	Flask / FastAPI
Frontend	React / HTML CSS
Database	MySQL / MongoDB
Deployment	Docker / Cloud Platform
Weather API	OpenWeather API

Expected Outcome

The proposed system is expected to help farmers identify diseases earlier and take preventive actions before severe crop damage occurs. By combining image analysis with environmental intelligence, the system can provide a more reliable and practical agricultural support solution. The project may contribute to improved crop management, reduced agricultural loss, and smarter farming practices.

Conclusion

This project presents a prototype concept for an AI-based paddy disease detection and forecasting system. The proposed system combines crop image analysis, weather forecasting, soil analysis, and crop health monitoring to create a complete agricultural decision support platform.

Instead of relying only on visible symptoms, the system aims to understand the environmental and biological factors responsible for disease spread. By predicting future disease risks and generating preventive suggestions, the system can support farmers in making better agricultural decisions and reducing crop loss.

Human civilization invented neural networks, cloud computing, and predictive intelligence, and finally arrived at the conclusion that rice plants also require software support. Technically absurd. Practically useful.